Section rotation to obtain across and along section velocities: How to?

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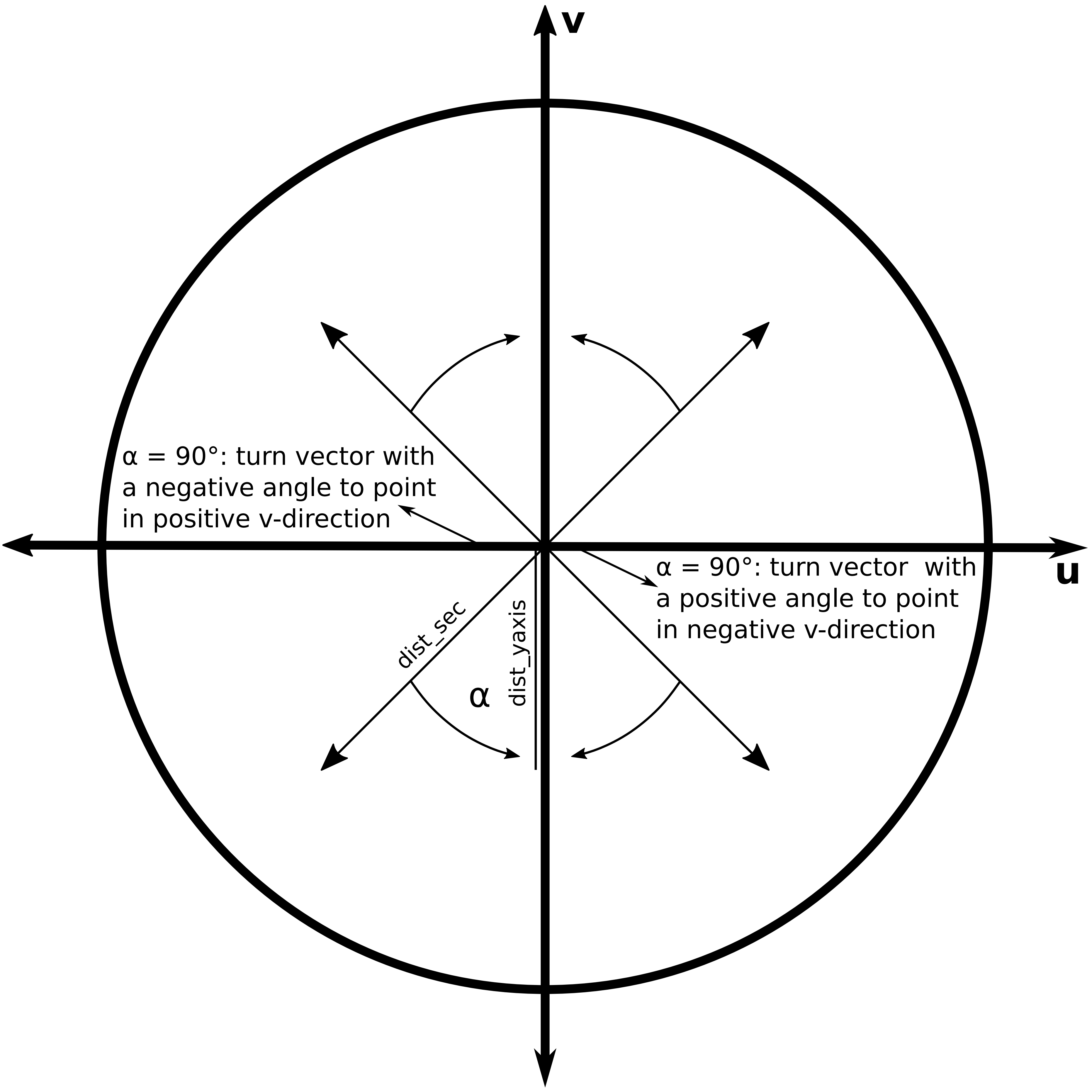

This sheet explains the different steps to rotate a section with given u and v velocities to obtain velocities across and along the section with example code in MATLAB:

1) Determine the length of the section using the SeaWater-Routine sw_dist:
dist_sec = sw_dist([lat(1) lat(2)], [lon(1) lon(2)], 'km');

2) Determine the length of the y-axis spanned by the section
dist_yaxis = sw_dist([lat(1) lat(2)], [lon(1) lon(1)], 'km');

3) Determine angle spanned by the section
α = acosd(dist_yaxis/dist_sec);

4) Depending on the orientation of the section, it is necessary to either turn it in positive (counter-clockwise) or negative direction.
Section that are pointing northward are turned northward, sections that are pointing southward are turned southward.
Use the function agle_turn_kw returns the angle with correct sign.
α = angle_turn_kw(lon_sec,lat_sec,α);
with lon_sec and lat_sec being the start and end points of the section.

5) Finally rotate the section using the function uvrot_kw:
[acrossvel, alongvel] = uvrot_kw(uvel, vvel, α);
with the u and v velocities uvel and vvel, respectively. Giving the velocities in across and along the section.

```
function [angle]=angle_turn_kw(cmems_lon_sec,cmems_lat_sec,angle)

if cmems_lon_sec(1) < cmems_lon_sec(length(cmems_lon_sec))
    if cmems_lat_sec(1) <= cmems_lat_sec(length(cmems_lat_sec))
        %the angle needs to be positive
    else
        %the angle needs to be negative
        angle = angle * (-1);
    end
else
    if cmems_lat_sec(1) <= cmems_lat_sec(length(cmems_lat_sec))
        %the angle needs to be negative
        angle = angle * (-1);
    else
        %the angle needs to be positive
    end
end

%Make sure the angle is no imaginary number
angle = real(angle);
```

```
function [ru,rv]=uvrot_kw(u,v,ang)
ang=ang*pi/180;
ru=u.*cos(ang)-v.*sin(ang);
rv=u.*sin(ang)+v.*cos(ang);
```